

SUPERIOR BONG, WI, USA

WMO: 726427

Lat: 46.689N

Lon: 92.094W

Elev: 674

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-9.3	-21.9	1.7	-13.1	-17.9	2.1	-7.9	23.0	17.4	20.1	17.3	6.0	230	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	86.2	72.0	82.5	69.3	80.7	67.9	75.0	83.0	72.4	80.0	69.9	76.7	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	122.9	80.3	70.1	113.5	77.2	67.5	103.7	74.4	38.9	83.3	36.5	79.6	34.4	75.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.7	16.7		DB	-20.4	92.3	5.5	2.8	-24.4	94.3	-27.6	95.9	-30.7	97.5	-34.7	99.5
				WB	-20.8	78.3	5.4	2.7	-24.7	80.2	-27.9	81.8	-30.9	83.3	-34.9	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	13.9	16.5	27.9	38.8	48.7	57.7	66.6	64.9	57.0	44.9	32.5	19.1	
	DBStd	20.20	12.83	11.57	11.20	7.26	7.51	7.42	6.14	5.81	7.43	7.53	9.87	11.65	
	HDD50	4924	1120	938	685	345	118	15	0	0	19	198	527	959	
	HDD65	9032	1585	1358	1149	786	507	238	57	73	257	624	974	1424	
	CDD50	1581	0	0	1	9	78	246	513	462	230	39	3	0	
	CDD65	215	0	0	0	0	3	19	105	70	17	1	0	0	
	CDH74	2523	0	0	0	8	113	347	1144	696	195	20	0	0	
	CDH80	686	0	0	0	2	27	83	345	185	41	3	0	0	
Wind	WSAvg	6.8	7.0	7.2	7.0	8.3	7.6	6.4	6.0	5.5	6.1	6.9	7.1	6.8	
Precipitation	PrecAvg	30.5	0.9	0.9	1.4	2.6	3.1	4.2	4.4	3.7	3.7	3.2	1.8	1.1	
	PrecMax	39.0	2.1	2.1	3.1	8.7	5.6	6.7	10.0	7.6	7.3	7.0	4.5	2.3	
	PrecMin	21.1	0.1	0.2	0.4	0.6	1.1	1.4	1.8	1.3	1.0	0.7	0.0	0.2	
	PrecStd	4.5	0.6	0.5	0.7	1.8	1.1	1.5	2.0	1.7	1.7	1.6	1.3	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	46.2	61.0	72.4	83.5	87.8	90.8	89.9	84.3	76.3	63.9	44.5	
		MCWB	38.7	38.9	48.4	54.2	64.1	73.9	74.9	74.5	70.5	60.9	52.1	38.3	
	2%	DB	37.3	39.9	52.5	63.9	76.2	81.8	87.6	84.4	79.9	67.6	55.2	39.2	
		MCWB	34.4	35.0	44.0	49.8	59.3	67.1	73.7	71.5	67.5	56.9	47.2	36.7	
	5%	DB	34.5	36.4	45.9	57.1	70.1	78.6	83.8	81.4	74.8	62.9	50.5	36.7	
		MCWB	32.9	33.3	39.4	45.5	56.5	64.3	70.7	69.3	64.6	54.3	44.6	35.0	
	10%	DB	31.9	33.6	41.5	51.9	63.9	73.1	81.1	77.9	70.7	57.2	46.2	34.6	
		MCWB	30.4	31.3	36.7	43.8	53.1	62.1	69.1	67.3	62.9	50.6	42.5	33.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	40.8	51.6	57.4	66.5	75.5	78.3	77.9	74.1	63.8	55.2	39.8
			MCDB	43.7	46.5	59.2	67.5	79.5	84.7	86.4	85.2	82.2	71.7	62.5	41.7
2%		WB	35.4	36.5	44.5	51.4	61.7	69.9	75.5	74.1	70.0	59.2	49.0	37.2	
		MCDB	37.0	39.5	51.1	63.2	71.5	77.0	83.4	81.9	75.5	65.7	53.9	38.8	
5%		WB	33.1	33.8	40.6	47.2	58.0	66.4	73.0	71.6	66.8	55.0	45.9	35.3	
		MCDB	34.7	36.2	45.3	55.9	67.7	74.8	81.0	78.6	73.0	60.4	49.0	36.6	
10%		WB	30.2	31.0	37.5	43.7	54.4	63.7	70.5	69.1	64.0	52.1	42.9	33.3	
		MCDB	31.6	33.0	40.9	50.6	62.7	70.9	77.9	75.2	68.7	56.3	46.3	34.6	
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.6	17.9	17.1	20.4	20.8	21.5	21.4	20.8	18.1	15.6	14.3	
		MCDBR	16.2	20.3	23.5	30.9	33.5	30.7	26.8	26.8	27.3	27.5	23.5	15.3	
		MCWBR	13.6	15.5	15.9	18.1	18.4	17.1	14.4	14.7	16.1	17.1	15.7	12.7	
	5% WB	MCDBR	15.7	17.6	22.3	28.9	29.5	27.1	25.2	23.9	23.3	23.4	21.1	13.6	
		MCWBR	13.4	13.8	15.4	17.9	18.4	17.1	15.3	14.7	15.2	16.0	15.1	11.8	
Clear-Sky Solar Irradiance	taub	0.249	0.272	0.315	0.355	0.385	0.381	0.386	0.392	0.369	0.317	0.291	0.269		
	taud	2.412	2.361	2.357	2.364	2.331	2.371	2.380	2.385	2.420	2.552	2.514	2.419		
	Ebn at Noon	275	289	289	286	279	280	277	271	267	270	256	252		
	Edh at Noon	21	28	33	36	39	37	37	35	31	23	20	19		
All-Sky Solar Radiation	RadAvg	461	806	1172	1458	1689	1831	1942	1625	1224	745	446	328		
	RadStd	49	73	109	146	125	126	78	117	83	95	51	33		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page